VAIBHAV PATEL

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PROFESSIONAL SUMMARY

Backend Engineer and Tech Lead with 3+ years shipping production systems at scale. Currently leading a 3-person team at Air Canada, owning 23 service repositories across a mission-critical AWS serverless stack. Brought flight disruption identification in-house, eliminating a hard external dependency and ensuring consistent handling across 21 booking channels. Experienced in cross-team coordination, disaster recovery, and driving features end-to-end from requirements to deployment. Proficient in TypeScript, Node.js, Go, Python, and AWS infrastructure (Lambda, AppSync, DynamoDB, CDK, Step Functions).

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Go, Java, SQL
Frameworks: Node.js/Express, React, Next.js, React Native, Laravel, Flask, Redux Toolkit
AWS: Lambda, AppSync, API Gateway, DynamoDB, Step Functions, SQS, SNS, CDK/CloudFormation, Cognito, Bedrock, S3, IAM
Databases: PostgreSQL, pgvector, Redis, Firebase/Firestore, MongoDB, MySQL
Tools & Integrations: OpenAI API, GetStream, Stripe, PayPal SDK, Docker, Terraform, GitHub Actions, Dynatrace, Splunk

EXPERIENCE

Backend Developer & Tech Lead — Jun. 2025 – Present

Air Canada (via TCS) — *Remote*

- Lead a 3-person backend team owning **23 service repositories**. Translate business requirements into technical tasks, manage backlog prioritization, and allocate team bandwidth across squads
- Brought flight disruption identification logic **entirely in-house**, eliminating a hard dependency on 1A (Amadeus). System handles standard and disrupted PNR flows across **21 booking channels**, covering e-Upgrades, duplicate segments, incomplete GDS data, and mid-journey disruptions
- Led **disaster recovery** for team-owned services: migrated repositories to a secondary AWS region, managed cross-account deployments, and coordinated with dependent teams to meet DR compliance
- Contributed to the **corporate booking flow** and coordinated cross-timezone delivery of the meal pre-order feature with end-to-end observability via **Dynatrace** and **Splunk**

Full Stack Developer — Mar. 2024 – Jun. 2025

Wiltekk Digital Solutions — *Remote*

Project 1: PowHerFit

- Developed robust web applications using **Laravel** framework, leveraging **MVC** architecture, eloquent **ORM**, and blade templating engine to deliver efficient and scalable solutions
- Implemented staging environment on **AWS**, reducing production deployment risks by enabling thorough API and CMS testing before code promotion
- Configured **Nginx** servers on **EC2** instances, resulting in optimized performance and enhanced communication between backend services and frontend interfaces, improving overall application responsiveness
- Established CI/CD pipelines using **Github Actions**, automating the build, test, and deployment processes, resulting in a **40%** reduction in deployment time and a decrease in production incidents

Project 2: Island Runners

- Built a two-sided ride-sharing marketplace for the Bahamas with separate React Native apps for customers and drivers. Implemented **PayPal** payments, real-time GPS tracking with **geofirestore**, Google Maps routing, and push notifications via **Firebase Cloud Messaging**
- Resolved **AWS Lambda** API bottlenecks achieving **10%** performance improvement and established **CI/CD** pipelines with GitHub Actions, cutting deployment time by **40%**

Cloud Engineer Co-op — Sep. 2023 – Dec. 2023

Manulife — *Toronto, ON*

- Achieved a significant reduction in subscription expenses by automating the deletion of expired and unused resources through the development of **Python**-based automation script, utilizing **Azure Automation, Azure Resource Graph**

- Automated provisioning of Azure resources through **Terraform**, reducing deployment time by **10%** and increasing scalability and reliability for clients
- Contributed to the improvement of code quality through active participation in code reviews, ensuring adherence to best practices and enhancing overall team efficiency

Software Developer (Part-Time) Jan. 2023 – Aug. 2023

Gravioty Research *Remote*

- Developed a multi-currency cryptocurrency wallet from scratch, providing support for Ethereum and Bitcoin blockchains using **React, Web3.js, Ethers.js, and Bitcoinjs-lib**
- Designed and implemented a robust state management architecture for the wallet utilizing **Redux**, resulting in optimized data handling
- Integrated **WebSocket** for real-time updates on cryptocurrency prices, ensuring users have access to the most up-to-date information
- Increased cross-browser compatibility by devising dynamic and easily configurable **webpack** scripts, ensuring seamless user experience across different browsers

Projects

Insights | *React 19, Node.js/Express, TypeScript, PostgreSQL, Redis, Firebase, Vite*

- Built full-stack financial intelligence platform enabling users to track multi-account banking, manage investment portfolios, create budgets, and receive AI-powered financial recommendations. Architected with **React 19 + Redux Toolkit** frontend and **Firebase Cloud Functions** backend
- Implemented sophisticated **AI transaction categorization** engine using **OpenAI embeddings** for semantic merchant classification, with **user preference override** support. Built **streaming chat interface** with **OpenAI API** for interactive financial queries, budget creation, and investment analysis
- Designed intelligent caching strategy with **Redis** and **Upstash**, reducing API response times by **50%**. Implemented embedding cache for merchant lookups and user category preferences to eliminate redundant OpenAI calls
- Engineered comprehensive test suite with **160+ Jest tests** covering financial calculations, category classification, and budget logic. Integrated **CI/CD pipelines** and automated email parsing for seamless transaction import
- Designed rich interactive dashboard with real-time net worth tracking, multi-asset allocation visualization, cash flow analysis, and goal progress monitoring using **Tailwind CSS** and **DaisyUI**

OGSA | *Firebase, React, JavaScript, Stripe, GCP*

- Implemented role-based access control, enabling seamless management of permissions, resulting in enhanced security and efficient utilization of features by **500+ users**
- Developed a robust ticketing system allowing for the creation and management of event tickets, including early bird options, leading to a **5%** increase in ticket sales and revenue generation
- Utilized **Stripe API and Firebase Cloud Functions** for seamless payment processing, resulting in secure transactions and streamlined revenue management, with over **800+** successful payments processed

Leadership & Community Involvement

Founder & Event Director Jun. 2021 – Jan. 2024

Ontario Gujarati Students Association *Toronto, ON*

- Lead an independent cultural group, organizing, and hosting over **5 events** annually with more than **700 attendees**
- Established a strong online presence through social media platforms, resulting in a **30%** increase in event attendance and engagement
- Coordinated aspects such as venue selection, budgeting, and team coordination, to ensure flawless execution
- Managed a team of **10+** individuals, delegating tasks and responsibilities to ensure the smooth running of events

Education

Wilfrid Laurier University Waterloo, ON

Honours Bachelor of Science in Computer Science *Sep. 2019 – Dec. 2023*

- Awarded Dean's List (2020 - 2021)

Certifications

Microsoft Certified: Azure Fundamentals Mar. 2024

Microsoft